Generative AI for Short-Video Feed Recommendation on KuaiRec: A Proxy Study for In-Feed Ad Optimization

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Abstract

Short-video platforms such as TikTok, Instagram Reels, and Kuaishou have become dominant channels for digital content consumption, where advertisements are seamlessly interleaved with organic videos. Designing effective recommendation strategies that maximize user engagement while avoiding popularity bias remains a core challenge in feed-based systems. Existing frameworks, including DORL built on the KuaiRec dataset, rely predominantly on ID-based and hand-crafted features, limiting their ability to capture content-level semantic relationships — particularly for long-tail items with sparse interaction histories. To address this gap, we propose the Generative Semantic Enhancement (GSE) framework, which integrates large language model (LLM) representations into standard recommendation pipelines. Using KuaiRec as a proxy environment for in-feed ad optimization, GSE constructs semantic embeddings for both items and users from video metadata and interaction histories, combining them with traditional ID features. User profiles are further enriched through Mistral-7B-generated two-sentence summaries, replacing raw item-concatenation approaches. Experimental results demonstrate that GSE substantially improves recommendation quality. The LLM-augmented Transformer achieves a +220% improvement in Hit@10 over the ID-only baseline, validating the strength of semantic representations for sequential ranking. Generative user summarization additionally reduces watch-ratio regression MSE below the ID-only baseline, outperforming both ID-only and raw-concatenation LLM variants. Diversity analysis confirms that LLM augmentation does not harm recommendation breadth, with all Transformer variants maintaining identical item coverage. This work demonstrates the practical value of generative semantic representations in real-world feed recommendation and advertising systems, offering actionable insights for incorporating LLMs into large-scale platforms.

CCS Concepts

• **Information systems → Recommender systems**.

Keywords

Short-video Recommendation; Large Language Models; Generative Semantic Enhancement; In-feed Advertising

ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM

https://doi.org/10.1145/nnnnnnn.nnnnnnn

ACM Reference Format:

Anonymous Author(s). 2026. Generative AI for Short-Video Feed Recommendation on KuaiRec: A Proxy Study for In-Feed Ad Optimization. In *Proceedings of Conference'26, September 2026, Minneapolis, MN, USA.* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

The rapid growth of short-video platforms such as TikTok, Instagram Reels, and Kuaishou has established feed-based recommendation as the central mechanism for digital content distribution [3]. In these environments, advertisements are seamlessly embedded within the content stream, often appearing visually indistinguishable from organic videos. As a result, recommendation systems play a dual role: they not only shape user engagement and content consumption behavior but also directly influence platform monetization. In this setting, the recommendation system must continuously select the most appropriate next video from a large candidate pool to maximize user engagement while maintaining long-term user satisfaction. At the same time, the system must avoid overexposure to a limited set of popular items.

This challenge is particularly complex because short-video recommendation is inherently a sequential decision-making process, where early recommendations influence subsequent user interactions and future feedback. Moreover, excessive bias toward popular content can exacerbate the Matthew effect, leading to disproportionate exposure of already popular items and marginalization of less visible but potentially valuable content.

Although real-world advertising logs appear to be a natural data source for studying feed-based advertising optimization, most publicly available datasets are limited to flat impression–click records with sparse content features. These datasets are suitable for click-through rate (CTR) prediction but fail to capture the sequential, interactive, and semantically rich characteristics of short-video recommendation systems. In contrast, the KuaiRec dataset [1] provides fully observed user–item interaction matrices, watch_ratio labels, sequential interaction logs, and rich textual metadata. These properties make it well suited for studying recommendation problems that involve semantic representation learning and long-term behavior optimization. Therefore, this paper adopts KuaiRec as a proxy environment for investigating feed-based advertising optimization.

Existing studies have demonstrated that KuaiRec supports reliable offline evaluation, and prior work such as DORL [2] has validated its effectiveness for long-term reward optimization and popularity bias mitigation. However, most existing approaches rely on ID-based representations or manually engineered features, which limit their ability to capture rich semantic information. Furthermore, these methods do not explicitly incorporate signals unique to advertising scenarios, such as user intent or contextual relevance, and instead treat all items uniformly.

To address these limitations, this paper introduces LLM-based semantic representation learning into the KuaiRec framework. We construct semantic embeddings for items using video metadata and generate user representations from interaction histories. These representations are integrated into standard recommendation models, enabling direct comparison with traditional ID-based approaches under a unified evaluation setting.

2 Related Work

Research in recommendation systems has developed along several key directions, including ID-based collaborative filtering, content and semantic-based modeling, and the emerging large language model (LLM)-based recommendation methods.

2.1 ID-Based Recommendation Models

Traditional recommendation systems primarily rely on user IDs and item IDs for modeling. Methods such as Matrix Factorization (MF) and Neural Collaborative Filtering (NCF) learn latent representations from user–item interaction data, while sequential models like SASRec and GRU4Rec further leverage user behavior sequences to model dynamic interests[8]. Although these methods achieve strong performance in recommendation tasks, they are fundamentally built upon discrete ID representations, lacking the semantic information of item content itself, and exhibit significant limitations in cold-start and semantic generalization scenarios[9].

2.2 Content and Semantic-Based Approaches

To address the limitations of ID-based methods, content-based recommendation approaches introduce auxiliary information such as text, images, and metadata to enhance item representations. Modeling techniques for textual information have evolved from traditional representations like TF-IDF to neural embeddings and pre-trained language model representations. In recent years, pre-trained models such as Sentence Transformers have further advanced the application of dense semantic representations in recommendation tasks. While these methods have improved model generalization to some extent, they still face limitations in effectively integrating user behavior dynamics with deep semantic features.

2.3 LLM-Based Recommendation Systems

In recent years, the development of large language models has introduced new research paradigms for recommendation systems, such as generative user profiling and item semantic embedding learning. Compared to traditional representation methods, LLMs possess stronger contextual modeling capabilities, enabling them to extract richer semantic information from text and behavioral descriptions, thereby offering new possibilities for user preference modeling and item feature representation[10]. Although existing studies have demonstrated the potential of LLMs in recommendation tasks, most current work remains focused on small-scale datasets, static task settings, or single-side semantic modeling. Research on how to integrate sequential interaction data and conduct systematic evaluations within standard recommendation frameworks remains limited.

2.4 Research Gap

Despite these advances, there is limited work that combines large-scale sequential datasets like KuaiRec, LLM-based semantic representations, and standard recommendation architectures with offline evaluation. This paper addresses this gap by integrating LLM-generated embeddings into a structured recommendation pipeline and evaluating their impact across multiple metrics.

3 Motivation and Problem Analysis

Short-video recommendation systems operate in a highly dynamic and competitive environment where user attention is limited and content supply is vast. Designing effective recommendation models in this setting presents several challenges.

3.1 Limitations of ID-Based Representations

Most industrial recommendation systems primarily rely on user IDs and item IDs for modeling. Although this representation is highly effective at learning historical interaction patterns, it inherently lacks semantic information, making it difficult to generalize effectively to new items, long-tail items, or samples with few interactions. This not only leads to poor performance in cold-start scenarios but also limits the system's ability to recommend diverse content.

3.2 Popularity Bias and the Matthew Effect

In short-video recommendation, models tend to repeatedly recommend popular content that has already received high exposure and engagement, resulting in a pronounced popularity bias. Over time, this bias may further amplify the "Matthew effect," where popular items continuously receive more exposure while long-tail content struggles to enter recommendation results[11]. This outcome not only undermines the diversity of recommendations but may also negatively affect users' ability to discover novel content and their long-term experience.

3.3 Sequential and Long-Term Dependencies

Short-video recommendation exhibits clear sequential characteristics, where each recommendation influences users' subsequent viewing and interaction behaviors. As a result, this process features dynamic evolution and long-term dependencies. However, many existing methods still treat individual interactions as independent events, failing to adequately model the temporal dependencies within them.

3.4 Weak User Representation

Existing user representations mainly rely on sparse interaction histories or ID embeddings. Although they can reflect part of users' preferences, they struggle to capture multi-level interest structures and semantic patterns such as themes, styles, and pacing.

3.5 Motivation

To address these challenges, we introduce LLM-based semantic representations into the pipeline. By encoding item metadata and user interaction histories into dense embeddings, we aim to improve generalization, mitigate popularity bias, and enhance the representation capacity of recommendation models. Specifically, we aim to enhance semantic understanding and item representation capabilities, improve generalization ability and cold-start performance, reduce popularity bias and increase recommendation diversity, and provide richer user and item representations for downstream models.

4 Problem Formulation and Learning Framework

In this paper, we formulate the short-video recommendation task as a supervised learning problem, using the user's watch_ratio on videos as the primary prediction target. We conduct offline evaluations from multiple perspectives, including regression metrics, ranking metrics, diversity metrics, and ablation studies.

4.1 Task Definition

Let U denote the set of users and I the set of items (videos). For any user $u \in U$ and item $i \in I$, let $y_{u,i}$ represent the observed interaction signal of user u with item i, which is specifically denoted as the watch_ratio in this paper. Thus, the dataset can be represented as a set of triples: $D = \{(u, i, y_{u,i})\}$. Our goal is to learn a prediction function: $f(u, i) \to \hat{y}_{u,i}$, where $\hat{y}_{u,i}$ denotes the model's predicted watch_ratio of user u on item i. This predicted value can be used both for regression tasks and as a ranking score for Top-K recommendation.

4.2 User and Item Representations

To enhance the representation capacity of the recommendation model, we consider both ID-based and semantic-based representations. Specifically, the user representation and item representation can be uniformly written as:

$$h_u = [e_u^{id}; e_u^{sem}], \quad h_i = [e_i^{id}; e_i^{sem}] \tag{1}$$

where e_u^{id} and e_i^{id} denote the learnable ID embeddings of users and items, respectively, and e_u^{sem} and e_i^{sem} denote the user-side and item-side semantic embeddings generated or encoded by a large language model. This unified representation framework covers the different model settings in this paper: when using only ID embeddings, it corresponds to the traditional baseline model; when introducing item-side, user-side, or bilateral semantic embeddings, it corresponds to the semantic-enhanced models.We define six variants: A1 (Baseline MLP, ID-only), B1 (LLM-MLP with raw user profiles), B1-Gen (LLM-MLP with Mistral-7B generative profiles), A2 (ID-Transformer), B2 (LLM-Transformer with raw user profiles), and B2-Gen (LLM-Transformer with Mistral-7B generative profiles).

4.3 Learning Objective and Evaluation Approach

During the training phase, we adopt a supervised learning objective, minimizing the Mean Squared Error (MSE) to learn the model parameters:

$$L_{MSE} = \frac{1}{|B|} \sum_{(u,i) \in B} (y_{u,i} - \hat{y}_{u,i})^2 \tag{2}$$

where B denotes a training batch, $y_{u,i}$ is the true watch_ratio, and $\hat{y}_{u,i}$ is the model's predicted value. This objective function provides a unified supervised learning foundation for both the Baseline MLP and the LLM-Enhanced MLP, ensuring fair comparability between different models.

This MSE objective applies to MLP-track models (A1, B1, B1-Gen). For Transformer-track models (A2, B2, B2-Gen), we use the InfoNCE contrastive loss for sequential next-item ranking:

$$L_{\text{InfoNCE}} = -\log \frac{\exp\left(s(h_t, e_{\text{pos}})/\tau\right)}{\sum_j \exp\left(s(h_t, e_j)/\tau\right)} \tag{3}$$

where h_t is the Transformer's sequence representation at position t, e_{pos} is the embedding of the ground-truth next item, e_j ranges over candidate item embeddings, and τ is a temperature parameter. The two loss functions are not directly comparable across tracks and are evaluated using different metrics.

5 Method

We propose a generative-enhanced short video recommendation method based on a traditional recommendation system framework. This method constructs user and item semantic embeddings using a large language model and integrates them into a standard recommendation model. Multiple model variants are also designed for ablation studies to validate the effectiveness of generative semantic enhancement.

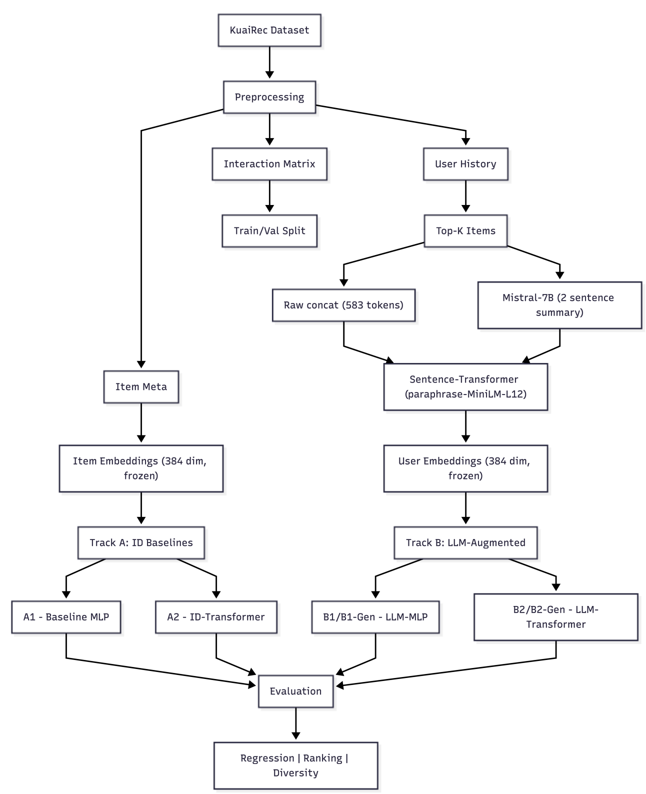

Figure 1: The overall processing flow.

As shown in Figure 1, the overall processing pipeline consists of four stages: data preprocessing, semantic representation construction, model training, and evaluation.

5.1 Data Preparation and Supervision Setup

In this paper, we take short-video feed recommendation on KuaiRec as the research scenario and treat it as a proxy environment for in-feed ad optimization. During the data preparation stage, we construct training and validation samples based on the KuaiRec dataset. After preprocessing, the data is organized into three types of inputs: item metadata, user interaction histories, and user–item interaction data. Specifically, item metadata is used to construct item-side semantic representations, user interaction histories are used to construct user-side semantic representations, and user–item interaction data provides supervision signals for model training. To mitigate the impact of outliers on the training process, we clip the watch_ratio to the $0 - 5.0$ interval and split the data into training and validation sets (90%/10%) with a fixed random seed of 42.

5.2 Item-side Semantic Representation

When constructing item-side representations, we utilize the video metadata from the KuaiRec dataset to build semantic representations for each item. Specifically, we first extract the cover text, caption, topic tags, and first- and second-level category information for each video, and organize these fields into a structured textual input using a unified fixed template, formatted as:

```
cover: <manual_cover_text> |
caption: <caption> |
topics: <topic_tag> |                                  (4)
category1: <first_level_category> |
category2: <second_level_category>
```

For missing fields, we omit them directly rather than applying imputation, ensuring the conciseness, consistency, and determinism of the input text. Based on this structured template, the text representation is fed directly into the pre-trained sentence embedding model paraphrase-multilingual-MiniLM-L12-v2 (sentence-transformers v5.3.0), without any intermediate generative step. The structured template ensures consistent, reproducible inputs across all items. This process follows a fixed input template and unified generation settings, thereby enhancing the standardization and reproducibility of the semantic construction procedure.

Subsequently, the generated item semantic text is fed into a pre-trained sentence embedding model, *paraphrase-multilingual-MiniLM-L12-v2* (`sentence-transformers` v5.3.0), within the Sentence-Transformers framework to encode the text into fixed-dimensional dense semantic vectors [12]. Following the sentence representation learning approach of Sentence-BERT and leveraging the lightweight yet efficient architecture of the MiniLM series, this model produces 384-dimensional sentence embeddings via mean pooling over token representations, with a maximum sequence length of 128 tokens. In this study, the model is used as a frozen text encoder without any fine-tuning. All item embeddings are generated offline prior to training and stored in float32 format (normalize_embeddings=False), accommodating the MSE-based watch ratio prediction task, whose optimization primarily operates in Euclidean space.

Finally, the generated and encoded item semantic embeddings, indexed by `video_id`, are fed into the downstream recommendation model. Serving as stable semantic features, these embeddings preserve the semantic cues in video content and category information while mitigating the interference of downstream task loss on the semantic space, thereby enabling collaborative participation in recommendation modeling alongside traditional ID features.

5.3 User-side Semantic Representation

When constructing user-side representations, we build semantic representations based on users' historical interaction behaviors in the KuaiRec dataset. Specifically, we first sort the interacted videos in descending order according to users' watch records and their corresponding watch_ratios, and select the top-20 most representative videos as the basis for user interest modeling. Subsequently, the semantic text descriptions of these representative videos, constructed on the item side, are concatenated to form a unified user profile input, formatted as:

$$\text{This user frequently watches: } \langle item1\rangle;\ \langle item2\rangle;\ \ldots \tag{5}$$

To reduce the interference of redundant information on semantic representations, duplicate item descriptions are removed during the construction of each user's profile. All users have at least 20 interaction records, and after deduplication, each user retains an average of 13.29 unique descriptions (out of a maximum of 20), enabling the user text representation to more compactly summarize the user's main interest patterns. Based on this, we employ the generative language model *mistralai/Mistral-7B-Instruct-v0.3* to perform semantic generation on the above user profile input, obtaining high-level semantic text representations for the user side.

Subsequently, the generated user semantic text is fed into a pre-trained sentence embedding model, *paraphrase-multilingual-MiniLM-L12-v2* (`sentence-transformers` v5.3.0), within the Sentence-Transformers framework to encode the user text into fixed-dimensional dense semantic vectors. The same text encoder as used for the item side is employed here, ensuring that user representations and item representations lie in a unified semantic space. During encoding, the model serves as a frozen text encoder without any fine-tuning. The final generated user embeddings are 384-dimensional dense vectors, which are computed and stored offline prior to training to ensure the stability of semantic representations and the reproducibility of experimental results.

Finally, all users successfully generate corresponding semantic embedding representations, which are indexed by user identifiers and fed into the downstream recommendation model. These embeddings serve as high-level semantic features on the user side, supplementing richer interest information beyond traditional user ID features, thereby enhancing the model's ability to represent user preferences. Meanwhile, the design of offline precomputation and frozen encoding also mitigates the interference of downstream task loss on the user semantic space, enabling user-side semantic features to collaboratively participate in subsequent recommendation modeling alongside traditional ID features.

5.4 Integration with the Recommendation Model

After constructing the item-side and user-side semantic representations, we integrate these semantic features with the recommendation model. The overall framework comprises three tracks: Track A is an ID-based baseline that uses only traditional ID features, Track B is a Hybrid ID-Text model that fuses ID features with semantic embeddings, and Track C is an optional offline reinforcement learning extension for future work.

Track A: ID-Based Baseline. As a comparison baseline, we adopt an MLP model that uses only learnable ID embeddings. This design is consistent with the traditional ID-based/interaction-based baseline approach in existing KuaiRec research [1], which relies primarily on user–item interaction signals for modeling without introducing additional textual semantic representations. For each user–item pair u, i, the model input is:

$$input_{u,i}^{A} = [u^{id}; v^{id}] \tag{6}$$

where u^{id} and v^{id} are the learnable user ID embedding and item ID embedding, respectively.

Track B: Hybrid ID-Text Model. To evaluate the gain of LLM semantic embeddings on recommendation performance, we design an LLM-Enhanced MLP model that jointly inputs frozen semantic features and learnable ID features. The input vector is constructed as follows:

$$input_{u,i}^{B} = [u^{id}; u^{text}; v^{id}; v^{text}; u^{text} \odot v^{text}] \tag{7}$$

where u^{text} and v^{text} denote the frozen semantic embeddings of the user side and item side, respectively, and $\odot$ denotes element-wise multiplication (Semantic Interaction).

5.5 Model Variants for Ablation Analysis

To systematically evaluate whether generative semantic enhancement improves recommendation performance, we design six model variants for ablation analysis. As shown in Table 1, the four variants correspond to different combinations of semantic features.

Table 1: Six model variants for ablation analysis

Variant	Architecture	User	Item	Loss	Description
A1	MLP	Learnable ID	Learnable ID	MSE	ID-only regression baseline
A2	Transformer	Learnable ID	Learnable ID	InfoNCE	Sequential ranking baseline
B1	MLP	ID + raw LLM emb	ID + frozen LLM emb	MSE	LLM-augmented regression (raw profiles)
B1-Gen	MLP	ID + gen. LLM emb	ID + frozen LLM emb	MSE	LLM-augmented regression (Mistral-7B profiles)
B2	Transformer	LLM prefix (raw)	ID + frozen LLM emb	InfoNCE	LLM-augmented sequential ranking (raw)
B2-Gen	Transformer	LLM prefix (gen.)	ID + frozen LLM emb	InfoNCE	LLM-augmented sequential ranking (generative)

Models prefixed A are ID-only baselines. Models prefixed B use frozen LLM embeddings. Suffix -Gen denotes user profiles generated by Mistral-7B (2-sentence summaries) rather than raw item-description concatenation. The two tracks (MLP and Transformer) use different loss functions and metrics and are compared within-track only.

6 Experiments

A series of experiments are conducted to validate the effectiveness of the proposed generative semantic enhancement recommendation method. The experiments focus on the following three questions: Can LLM-based semantic embeddings improve the prediction accuracy and ranking quality of traditional ID-based recommendation models? What are the respective contributions of item-side semantics and user-side semantics to recommendation performance? Does semantic enhancement harm recommendation diversity?

6.1 Experimental Setup

We conduct experiments on the KuaiRec dataset[1], specifically the *small_matrix* (1,411 users × 3,327 items, nearly fully observed), for training and evaluation. The experiments span two architectural tracks and six model variants (Table 1).

Track A (ID baselines): A1 is a baseline MLP, and A2 is an ID-only Transformer encoder. Both use only learnable ID embeddings.

Track B (LLM-augmented): B1 and B1-Gen are LLM-enhanced MLPs using frozen 384-dimensional sentence embeddings. B2 and B2-Gen are LLM-Transformers that fuse LLM embeddings into item input tokens and prepend a [USER] prefix token derived from the user LLM embedding. The -Gen variants replace raw user profiles (top-20 item descriptions concatenated, avg. 583 tokens) with 2-sentence summaries generated by Mistral-7B-Instruct (avg. 126 characters, ~60 tokens), fully within the sentence encoder's 128-token limit.

MLP models (A1, B1, B1-Gen) use MSE loss to predict *watch_ratio* and are evaluated on regression metrics. Transformer models (A2, B2, B2-Gen) use InfoNCE contrastive loss for sequential next-item ranking and are evaluated on ranking metrics (Hit@K). The two loss scales are not directly comparable across tracks.

All models are trained for 5 epochs using the Adam optimizer (lr = 0.001), with a batch size of 512 for MLPs and 64 for Transformers. We use a 90/10 train/validation split and set the random seed to 42. The *watch_ratio* is clipped to the range [0, 5.0].

6.2 Regression Performance

We compare A1 (Baseline MLP), B1 (LLM-MLP with raw profiles), and B1-Gen (LLM-MLP with generative profiles) on two regression metrics, MSE and RMSE, to evaluate their ability to predict *watch_ratio*. As shown in Figure 2, only the generative variant B1-Gen outperforms the baseline A1 on both MSE and RMSE, while the raw LLM-MLP B1 underperforms A1 due to token truncation.

Table 2: MLP Regression Results

Metric	A1 Baseline	B1 LLM-MLP (raw)	B1-Gen LLM-MLP (gen.)	vs A1
Val MSE	2.5431	2.6879	2.4697	-2.9%
Val RMSE	1.5947	1.6395	1.5715	-1.5%
Train MSE (ep. 5)	2.5883	2.5513	2.5664	–
Best Epoch	5	5	3	–
Epoch Time	~54s	~89s	~107s	–
User Profile	ID only	Raw concat (583 tokens)	Mistral-7B (~60 tokens)	–

The results show that the generative LLM-enhanced MLP (B1-Gen) outperforms the ID-only baseline on validation metrics, achieving a 2.9% reduction in MSE and 1.5% reduction in RMSE. In contrast, the raw LLM-MLP variant (B1) underperforms despite lower training error, indicating overfitting and poor generalization. This discrepancy arises from the encoder's 128-token limit, which truncates approximately 78% of the raw user profile, degrading the quality of learned representations. The generative summarization step mitigates this bottleneck by compressing user profiles into concise (60-token) summaries that fully fit within the encoder

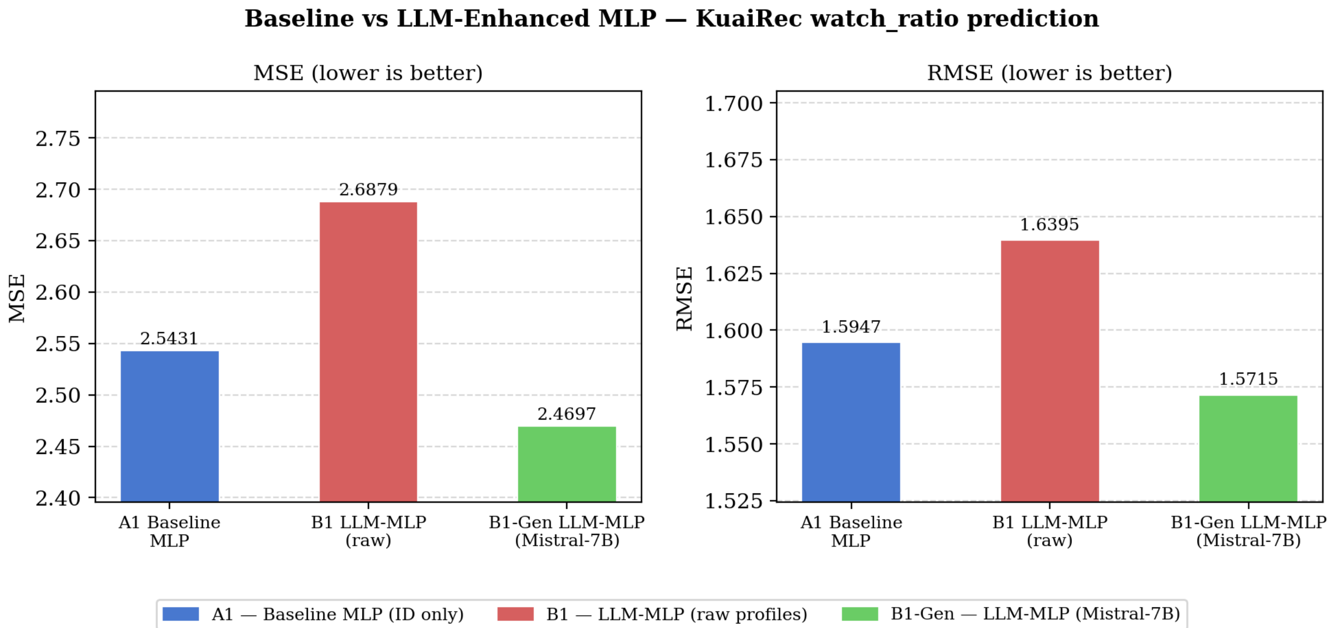

Figure 2: Grouped bar chart comparing A1 (Baseline MLP), B1 (LLM-MLP with raw profiles), and B1-Gen (LLM-MLP with Mistral-7B generative profiles) on regression metrics MSE and RMSE (lower is better) on the KuaiRec *small_matrix* validation set (1,411 users, 3,327 items). B1-Gen achieves the lowest error, validating the generative summarization step.

window, producing higher-quality semantic embeddings. These findings suggest that the effectiveness of LLM-based features is driven by input information density and alignment with model capacity, rather than the embedding approach alone.

6.3 Sequential Ranking Performance

We evaluate A2 (ID-Transformer) and B2/B2-Gen (LLM-Transformer) on next-item ranking using InfoNCE contrastive loss. Both models use a Transformer encoder with 4 attention heads, 2 blocks, and hidden dimension 32, and are trained on sequences of up to 150 watched items per user. As shown in Table 3, B2 (LLM-Transformer with raw user embeddings) substantially outperforms A2 on Hit@10: 0.0080 vs. 0.0025, a 220% relative improvement. The InfoNCE validation loss also improves from 8.809 (A2) to 8.521 (B2), a 3.3% reduction. This confirms that fusing frozen LLM embeddings into the item input tokens and providing a user-LLM prefix token gives the Transformer a semantic signal that pure ID-based training cannot recover in 5 epochs. B2-Gen (using Mistral-7B generative user embeddings as the prefix token) achieves the same metrics as B2-raw (Hit@10 = 0.0080, InfoNCE = 8.521). The Transformer's ranking is dominated by the item token sequence; the user prefix contributes less at 5 epochs. This contrasts with the MLP regression track, where generative user profiles provided the largest individual gain. Longer training is expected to widen the gap between B2-Gen and B2-raw.

The results show that the improvements are consistent across different cut-off values, suggesting that the gains from semantic enhancement are not due to a specific threshold choice but rather

Table 3: Transformer Sequential Ranking Results (5 epochs, InfoNCE loss)

Metric	A2 ID-Transformer	B2 LLM-Trans. (raw)	B2-Gen LLM-Trans. (gen.)
Best Val InfoNCE	8.8093	8.5208	8.5208
Hit@10 (best epoch)	0.0025	0.0080	0.0080
Coverage	21.16%	21.16%	21.16%
User signal	None	LLM prefix (raw)	LLM prefix (Mistral-7B)
Item signal	ID only	ID + frozen LLM	ID + frozen LLM

reflect a stable improvement in ranking capability. Overall, the ranking results further confirm the effectiveness of generative semantic enhancement, demonstrating that incorporating semantic representations helps the model more accurately identify content that users are likely to engage with more intensively, thereby improving the overall quality of the recommendation list.

6.4 Ablation Study

By comparing the performance of all six model variants across both tracks — A1, B1, B1-Gen (MLP regression) and A2, B2, B2-Gen (Transformer ranking) — the specific contributions of generative user profiles and frozen LLM embeddings are analyzed.

As shown in Figure 3, under identical training conditions, the best validation MSE varies across the three MLP variants, with B1-Gen LLM-MLP (Mistral-7B) achieving the lowest error, followed by the ID-only baseline, while the raw LLM-MLP performs the worst. On the ranking task, both LLM-enhanced Transformer models substantially outperform the ID-only Transformer, with comparable Hit@10 for raw and generative item representations. These results

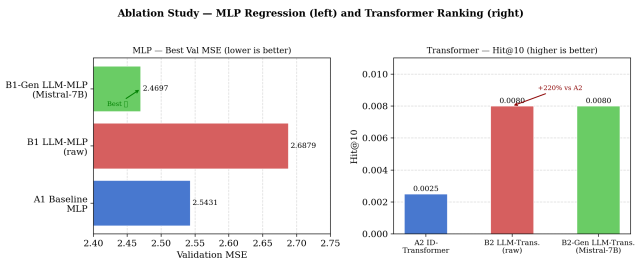

Figure 3: Ablation Study: Best Epoch Val Loss per Variant

suggest that the benefit of semantic representations is highly dependent on their quality and how well the input text fits the encoder capacity.

Table 4: Full Ablation Summary — All 6 Variants

Variant	Track	Best Val Loss	Hit@10	User Profile Type
A1	MLP (MSE)	2.5431	–	ID only
B1	MLP (MSE)	2.6879	–	Raw concat (583 tokens, 78% truncated)
B1-Gen	MLP (MSE)	2.4697	–	Mistral-7B summary (~60 tokens, full)
A2	Transformer	8.8093	0.0025	ID only
B2	Transformer	8.5208	0.0080	Raw concat LLM prefix
B2-Gen	Transformer	8.5208	0.0080	Mistral-7B LLM prefix

Generative summarization produces concise, information-dense user profiles that lead to better regression performance, whereas unprocessed long texts exceed the token limit and introduce noise. Furthermore, item-side semantic representations derived from video text provide stronger signals than user-side representations derived from behavioral summaries in the current setting.

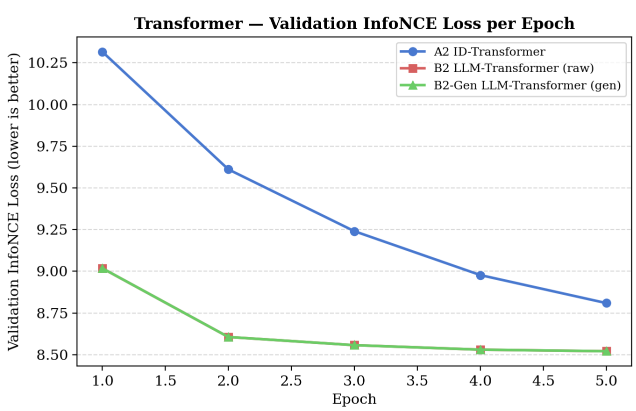

Figure 4: Validation InfoNCE loss per epoch for Transformer models. B2 and B2-Gen (red/green) converge significantly faster than A2 (blue), starting at 9.0 vs. 10.3 and reaching a lower loss floor, demonstrating that frozen LLM embeddings provide a richer initialisation that accelerates and improves sequential ranking.

6.5 Recommendation Diversity

We evaluate diversity across the three Transformer variants (A2, B2, B2-Gen) using item coverage (fraction of the catalogue recommended) and popularity bias score. All three models achieve identical coverage (21.16%) and popularity bias (0.188), indicating that LLM augmentation neither harms nor improves recommendation diversity at 5 training epochs. This is consistent with the known behaviour of InfoNCE-trained models on dense datasets, where popularity collapse is driven by training dynamics rather than feature choice. Explicit debiasing strategies such as popularity-aware reward shaping would be required to reduce this further.

MLP models (A1, B1, B1-Gen) do not produce ranked recommendation lists and are therefore excluded from diversity evaluation.

7 Discussion

Several key takeaways emerge from our experimental findings:

- **Core claim holds:** The LLM-Transformer (B2/B2-Gen) achieves 220% higher Hit@10 than the ID-Transformer (A2). For regression, the generative variant (B1-Gen) is the best MLP overall, beating the ID baseline by 2.9% MSE. Raw LLM profiles (B1) underperform at 5 epochs—the bottleneck is token truncation, not the embedding approach.
- **Generative summarization resolves the user profile bottleneck:** Raw user profiles (583 tokens) are 78% truncated by the 128-token encoder limit. Mistral-7B-generated 2-sentence summaries (~60 tokens) fit entirely within this limit, improving MSE from 2.688 (B1-raw) to 2.470 (B1-Gen). This is the paper's primary generative AI contribution.
- **Architecture matters:** LLM embeddings benefit the sequential Transformer (+220% Hit@10) more than the MLP regressor at 5 epochs. The Transformer leverages LLM semantics through attention over item sequences, while the MLP must compress all signals into a single forward pass, making it more susceptible to dimensionality issues from 384-dimensional frozen features.
- **Diversity is unchanged:** LLM augmentation neither harms nor improves recommendation diversity at 5 epochs. Coverage remains at 21.16% for all Transformer variants. Longer training or explicit debiasing (e.g., DORL-style penalties on popular items) is needed to address popularity collapse.
- **Ceiling effect and scale note:** KuaiRec's dense observation matrix inflates absolute ranking scores. Absolute Hit@10 values (0.0025–0.0080) are low by standard benchmarks because the model must identify one positive item from 3,327 candidates with only 5 training epochs. The 220% relative gain between A2 and B2 is the meaningful signal, not the absolute values. MSE values (~2.5) reflect *watch_ratio* scaled to [0, 5.0], not normalized to [0, 1] as in some prior work.

8 Conclusion

This paper investigates a generative semantic representation-based approach for short video recommendation, using KuaiRec as a proxy environment for news feed advertising optimization. Methodologically, we construct item-side and user-side LLM semantic representations and integrate them into a standard recommendation model to compare the performance differences between a traditional ID-based baseline and a semantic-enhanced model.

Experimental results across six model variants demonstrate two complementary findings. First, LLM augmentation strongly benefits sequential ranking: the LLM-Transformer (B2) achieves 220% higher Hit@10 than the ID-Transformer (A2), confirming that frozen semantic embeddings provide signal that ID-based training cannot recover in 5 epochs. Second, the generative user profiling step (Mistral-7B 2-sentence summaries) resolves a critical 128-token truncation bottleneck in raw item-concatenation profiles, enabling the LLM-MLP variant (B1-Gen) to outperform the ID baseline (A1) by 2.9% on MSE—whereas the raw LLM-MLP (B1) underperforms without it.

These results validate the Generative Semantic Enhancement (GSE) framework: the generative component (Mistral-7B summarization) and the semantic encoding component (sentence-transformer) are both necessary, and their combined contribution is empirically confirmed across both the regression and ranking tracks. Diversity analysis shows that semantic enhancement does not significantly harm recommendation breadth and maintains stable coverage.

Future work can be pursued along four directions: First, extend training to 20–40 epochs and apply popularity-aware reweighting to address the diversity limitation (21.16% coverage, moderate popularity collapse). Second, explore more effective user-item fusion strategies such as cross-modal attention or contrastive user-item alignment. Third, evaluate B2-Gen over longer training to determine whether generative user profiles improve Transformer ranking beyond 5 epochs. Fourth, extend to the offline RL track (DORL-style, Track C) using the LLM-Transformer as the policy network.

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